Women Safety Route App

INTERNSHIP PROJECT REPORT

Submitted in partial fulfillment of the requirements for the award of the degree

Of

BACHELOR of TECHNOLOGY
DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCES

By

Jahanvi Sharma
08001172024

Guided by

Dr Ritu
Research Associate,
Anveshan Foundation ,IGDTUW

INDIRA GANDHI DELHI TECHNICAL UNIVERSITY FOR WOMEN

NEW DELHI – 110006

2 june-11 july 2025

CERTIFICATE

I, Jahanvi Sharma, certify that the Internship Project Report entitled "Women Safety Route App" is done by me and it is authentic work carried out by me at Anveshan Foundation, IGDTUW. For this project, no work has been submitted before for any degree or diploma of the award, to the best of my knowledge and belief.

Signature of the student 

Certified that the project report entitled "Women Safety Route App" done by the above student is completed under my guidance.

Signature of Mentor 

Mentor Name: Dr Ritu
Designation: Research associate
Compnay/Univeristy Name:
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ACKNOWLEDGEMENT

I would like to acknowledge my mentor Dr Ritu for his very helpfulcomments, support and encouragement.

Finally, I am grateful to Anveshan Foundation, IGDTUW for providing a healthy, supportive and understanding environment. They allowed me the freedom to explore innovative models to simplify a complex business problem. This made my project work possible without any hindrance.

Signature of the Student

Jahanvi Sharma

08001172024

DECLARATION

I, **Jahanvi Sharma**, solemnly declare that the internship project report, **Women Safety route app,**is based on my own work carried out under thesupervision of **Dr. Ritu**. I assert the statements made and conclusions drawn are an outcome of my work. I further certify that:

I. The work contained in the report is original and has been done by me under the supervision of my supervisor.

II. The work has not been submitted to any other Institution for any other degree/diploma/certificate in this university or any other University of India or abroad.

III. We have followed the guidelines provided by the university in writing the report.

IV. Whenever we have used materials (text, data, theoretical analysis/equations, codes/program, figures, tables, pictures, text etc.) from other sources, we have given due credit to them in the report and have also given their details in the references.

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Jahanvi Sharma

08001172024

Indira Gandhi Delhi Technical University for Women

Internship Offer Letter

jahanvi jahanvi <jahanvi080btcseai24@igdtuw.ac.in>

6 Weeks Summer Internship on PYML 2025 - Anveshan Foundation IGDTUW, WhatsApp Group Link

2 messages

COE- AI IGDTUW <coeai@igdtuw.ac.in> Sun, May 25, 2025 at 8:07 PM
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Bcc: jahanvi080btcseai24@igdtuw.ac.in

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CERTIFICATE OF COMPLETION

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For successfully completing the Six Weeks Summer Internship on "PYTHON & MACHINE LEARNING" from 2nd June - 11th July, 2025 conducted by Anveshan Foundation, IGDTUW.

Dr. Rahul Sachdeva
Faculty Coordinator
Anveshan Foundation

Prof. Arun Sharma
Dean - Academic Affairs
IGDTUW

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About Company

Anveshan Foundation, IGDTUW is an incubation and innovation foundation which supports students and new startups working on new technologies and creative ideas . Students can develop skills , related to entrepreneurship,
research and technical innovation under the anveshan foundation . The major focus is on promoting women in fields of technology and innovation . It functions as a Section 8 company since October 2016 and is recognized
by the Department of Science & Technology (DST), Government of India, as a Technical Business Incubator (TBI).
As part of this initiative, various internships and training programs are conducted in domains
like Python, Machine Learning, Artificial Intelligence, and Data Science to help students gain practical exposure.

My internship experience under this initiative helped me gain hands-on exposure to Python

and Machine Learning while developing a better understanding of how real-world applications are built.

The foundation's initiative helped me bridge the gap between theoretical knowledge and its practical applications, giving me confidence to work on real-world problems.

Objectives of the Internship

To independently explore and learn advanced libraries like GeoPandas and Folium for data visualization and mapping applications.

To develop logical thinking and problem-solving skills through coding exercises and project-based learning.

To understand the process of applying theoretical concepts of AI and ML to practical scenarios.
To learn python programming language and its real-world applications.

To learn the fundamentals of Machine Learning and understand how the real world models are built and trained.

To get hands-on experience with Python libraries such as NumPy, Pandas, Matplotlib, and Scikit-learn for data handling and model implementation.

Timeline :

Week 1: Python Basics and Data Preprocessing

- Learn Python syntax, data types, conditional statements, loops, list comprehensions, string and dictionary operations.
- Introduction to working with external libraries (NumPy, Pandas).
- Data cleaning, handling missing values, and scaling/normalization.

Week 2: Data Science Libraries and Data Visualization

- Deep dive into NumPy and Pandas for data structures, indexing, manipulation, grouping, merging.
- Use Matplotlib and Seaborn for data visualization (line charts, bar charts, scatter plots).
- Begin simple data exploration projects.

Week 3: Advanced Data Preprocessing and Introduction to Machine Learning

- Handle missing data, parse dates, correct inconsistent entries.
- Study basic supervised learning algorithms, categorical variable handling, and build the first ML models using scikit-learn.

Week 4: Model Building and Validation

- Explore multiple ML algorithms (classification and regression—Decision Tree, Linear Regression, etc.).
- Learn about pipelines, model validation, system optimization, and cross-validation techniques.
- Practical implementation and evaluation of ML models using real-world datasets.

Week 5: Advanced ML Concepts and Applications

- Study ensemble methods (XGBoost, CatBoost, LightGBM), underfitting/overfitting, Principal Component Analysis (PCA), and data leakage.
- Work on mini-projects involving classification, regression, and web scraping problems.

Week 6: Capstone Project and Reporting

- Undertake a complete data science or machine learning project independently or in teams.
- Prepare thorough documentation of the work, create visualizations, and practice presenting findings. Final submission includes code, reports, and presentations.

Tasks undertaken

1. Research Problem Finalization (Deadline: 17th July 2025)
The initial phase of the project focused on identifying and defining the core research problem. A preliminary background study and literature review were conducted to gain a deeper understanding of the topic. Based on this analysis, the final problem statement was formulated and recorded in the project tracking sheet for faculty review and approval.

2. Synopsis Submission (Deadline: 21st July 2025)
In the next phase, a concise synopsis (1–2 pages) was prepared, summarizing the project's objectives, proposed methodology, dataset information, tools, and libraries to be utilized. The synopsis acted as a foundational document, outlining the project plan and guiding the development process ahead.

3. Project Completion and Code Submission (Deadline: 15th August 2025)
This stage involved the complete design, development, and testing of the proposed system or model. The final code was implemented, debugged, and properly documented with clear comments to ensure ease of understanding, reproducibility, and future reference.

4. Research Paper Writing and Submission (Deadline: 30th August 2025)
The concluding task focused on writing and submitting a comprehensive research paper summarizing the entire project. The paper included the introduction, literature review, methodology, experimental analysis, results, discussions, and conclusion. It also emphasized the major findings, outcomes, and insights derived from the research work.

Work Description

The project titled **"Women's Safety Route App"** aims to identify and visualize the safest travel routes within Delhi based on real-world crime data. The system calculates area-wise safety scores using geospatial analysis and graph-based algorithms, providing users with an interactive map to find the safest path between two locations. The work was carried out in the following major stages:

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1. Data Collection and Integration

Multiple open-source datasets were collected from Kaggle and other publicly available sources to support the project's objectives:

- **Crime Dataset:** A JSON file containing attributes such as *crime_number, crime_type, longitude, latitude, occurred,* and *disposition.* This dataset was crucial for calculating safety scores and generating the Delhi safety heatmap.
- **Delhi Boundary Dataset:** A GeoJSON file containing Delhi's administrative boundary coordinates (*type, name, crs, features*). It was used to plot and visualize the Delhi map accurately.
- **Delhi Police Stations Dataset:** A KML file containing *police station names* and their corresponding *longitude* and *latitude.* It enabled marking police station locations on the safety map.
- **Delhi Metro Dataset:** An Excel file containing *station ID, names, distance, metro line, longitude, latitude, layout,* and *opened year.* It was merged with the crime dataset to provide a broader spatial reference for mapping.

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2. Libraries and Tools Used

The implementation utilized a combination of Python libraries for data handling, mapping, and visualization:

- **Data Handling & Processing:** pandas, numpy, json, xml.etree.ElementTree, base64, io
- **Geospatial & Mapping:** geopandas, shapely, folium, streamlit-folium
- **Graph & Routing:** NetworkX
- **Visualization:** matplotlib
- **Web Application:** streamlit, webbrowser

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3. Data Preprocessing

The raw datasets were preprocessed for consistency and integration:

- The *crime dataset* was loaded and formatted; longitude and latitude columns were standardized and rounded to two decimal places.
- The *Delhi metro dataset* was cleaned similarly, and both datasets were merged to create a comprehensive dataframe for geospatial analysis.

- The *police station dataset* (KML format) was converted to CSV, coordinates were split and cleaned, and unnecessary spaces were removed.
- Dataframes were created for all cleaned datasets to enable efficient data manipulation.

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4. Grid Creation and Boundary Overlay

The *Delhi boundary GeoJSON* file was loaded and converted into a coordinate reference system (in meters). Using its geographical bounds, grid cells (polygons) were created to divide the city into smaller spatial segments.
These grids were overlaid on the Delhi boundary using GeoPandas, and the merged dataframe containing crime locations was spatially joined to the respective grid cells to determine which crimes occurred within each cell.

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5. Safety Score Calculation

For each grid cell, the total number of crimes was computed to estimate its relative safety level.
Safety scores were normalized between 0 and 1 using the formula:

Safety Score=1−Crime CountMax Crime Count\text{Safety Score} = 1 - \frac{\text{Crime Count}}{\text{Max Crime Count}}Safety Score=1−Max Crime CountCrime Count

Cells with zero crime were assigned a safety score of 0. Each grid cell's geometry and safety score were then used to build a graph structure for route computation.

Edge weights between adjacent cells were determined using the formula:

Weight=1−(Safety Score1+Safety Score2)2\text{Weight} = 1 - \frac{(\text{Safety Score}_1 + \text{Safety Score}_2)}{2}Weight=1−2(Safety Score1+Safety Score2)

This ensured that routes passing through safer cells had lower cumulative weights.

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6. Pathfinding and Route Optimization

A *nearest_node* function was implemented to map user-provided start and destination coordinates to their closest grid cells.
The **Dijkstra algorithm**, implemented using the NetworkX library, was applied to identify the safest route between these two nodes.
The resulting route was visualized on a Folium map, highlighting the safest path across Delhi.

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7. Geospatial Mapping Workflow

The project incorporated multiple geospatial layers:

- The Delhi boundary was plotted using a GeoJSON layer for spatial reference.
- Safety grids were color-coded from dark green (safest) to red (most unsafe) based on safety scores.
- Start and destination points were represented with green and red markers, respectively.
- Police station locations were visualized using blue markers.
- The final map was saved as an HTML file and automatically opened in a web browser for user interaction.

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8. Visualization and User Interface

Several visualization modules were implemented to provide insights:

- **Crime Type Distribution:** A bar chart showing frequency of each crime type.
- **Crime Location Scatter Plot:** Longitude and latitude plotted as dots to represent crime occurrence points.
- **Safety Score Map:** A color-coded safety heatmap with start, end, and safest route overlays.
- **Police Station Map:** A folium-based map displaying all police stations as blue popups.

A Streamlit-based GUI allowed users to input start and destination coordinates, visualize the safest route, and explore safety insights interactively.

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9. Results

The final output was a **functional web interface** where users can enter their source and destination coordinates to view the safest possible route in Delhi, along with detailed safety visualizations and police station locations.

Work Outcome

The project titled **"Women's Safety Route Finder using Crime Data Analysis"** successfully demonstrated how geospatial mapping, data analytics, and graph algorithms can be integrated to identify and visualize the safest travel routes across Delhi. Key outcomes include:

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1. Performance and Implementation Results

- A **500m spatial grid** was generated over the Delhi map using the Delhi Boundary GeoJSON dataset, enabling localized safety analysis.
- Each grid cell was assigned a **Safety Score**, calculated by normalizing crime counts derived from the Kaggle crime dataset, with values ranging from 0 (least safe) to 1 (most safe).
- The **Dijkstra's Algorithm** was applied on the constructed weighted graph (where nodes represented grid cells and edges represented adjacency with safety-based weights) to determine the **safest route** between user-defined start and destination points.
- **Police station locations** were successfully marked as blue pop-ups, providing an additional layer of safety information on the Delhi map.

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2. Visualization and Insights

- Multiple **data visualizations** were generated using *Matplotlib* and *Folium*:
  - **Bar Graph** showing the frequency of different types of crimes.
  - **Scatter Plot** displaying the spatial distribution of crime occurrences.
  - **Safety Score Heatmap** illustrating the safety levels of Delhi regions, with color coding from red (unsafe) to green (safe).
  - **Safest Route Map** highlighting the safest travel path in blue, with start and end points marked in green and red respectively.
  - **Police Station Map** showing all Delhi police stations as blue pop-ups for public safety reference.
- The visual outputs were exported as **interactive HTML maps**, enhancing accessibility and interpretability.

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3. Tools and Technologies

- Data handling and preprocessing were achieved using **Pandas**, **NumPy**, and **JSON/XML** libraries.
- Geospatial computations and visual mapping were performed using **GeoPandas**, **Shapely**, and **Folium**.
- **NetworkX** was employed for graph creation and shortest path computation.
- **Streamlit** and **Webbrowser** modules were integrated to enable a simple GUI for user interaction and map visualization.

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4. Final Deliverables

- A functional **Graph-based Route Safety Model** capable of computing and displaying the safest path across Delhi.
- **Interactive visual dashboards** representing crime data distributions and geospatial patterns.
- A **modular, documented Python codebase** with preprocessing, grid creation, path computation, and visualization modules.
- **HTML-based interactive safety maps** embedded with crime heatmaps and police station markers.

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5. Conclusion

The project effectively showcased the potential of combining **data analytics, geospatial intelligence, and graph algorithms** to enhance women's safety during travel. It serves as a prototype for real-world safety navigation systems. Future enhancements could include **real-time crime data integration**, **live GPS tracking**, and **expansion to other metropolitan cities**

CONCLUSION

The goal of the project is to find the safest route for women in order to make their travel safer .

This in my project was done by using crime dataset of delhi and overlaying a grid of 500m in size all

over delhi and assigning each grid a safety score and then calculating the safest route using dijkstra

algorithm . Police stations are shown as blue colored pop ups .and also comparative plots were made

for different types of crimes happening in delhi . The project can really contribute to being a safety tool

for women while travelling . in the future real time crime data can be integrated in the project and it can

also be explanded to other cities .

Plagiarism report

Jahanvi Sharma

Women Safety

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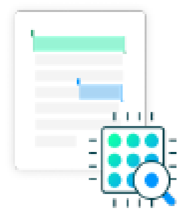

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Research Paper

Title: *Women Safety Route App – A Data-Driven Application for Safe Route Navigation in Delhi*
Author: Jahanvi Sharma
Conference: Submitted
Status: Completely Written

Research Description

This research presents the design and implementation of a data-driven application, **Women Safety Route App**, aimed at enabling safe navigation for women in Delhi using crime analytics and geospatial data. Recognizing that traditional navigation systems focus solely on the shortest route, this study introduces an algorithmic framework that prioritizes **route safety** as a core metric.
The proposed system integrates real-world crime datasets and spatial mapping techniques to calculate the safest possible travel path between two points. It employs **Python's geospatial and data science libraries**, and the **Dijkstra algorithm** to identify the optimal safe route, while also displaying nearby police stations to assist in emergencies.

This work addresses an urgent social and technological problem — empowering women with intelligent mobility tools that support **preventive safety** instead of post-incident response.

Objectives

The primary objectives of the research are to:

1. Develop a **data-driven safe routing system** for women that uses crime frequency and spatial data to calculate safety scores.
2. Integrate **geospatial mapping** and **graph algorithms** to identify the safest path rather than the shortest one.
3. Enhance situational awareness by visualizing **crime hotspots and police stations** on an interactive map.
4. Build a lightweight, user-friendly application using **Python Streamlit** that can operate efficiently on real-world datasets.

Dataset and Features

The system uses four key datasets:

- **Crime Dataset (Kaggle):**
  Contains detailed records of crimes in Delhi with fields such as *crime number, type, longitude, latitude,* and *date/time*.
- **Delhi Boundary Dataset (GeoJSON):**
  Defines the spatial extent of Delhi, used to plot the map boundaries.
- **Delhi Police Stations Dataset (KML):**
  Includes police station names and geolocations, displayed as pop-ups on the map.
- **Delhi Metro Dataset (CSV):**
  Adds broader spatial reference to connect route data with real-world urban points.

Each dataset is preprocessed using **Pandas** and **GeoPandas**, cleaned, and spatially aligned to generate a 500m × 500m grid overlay on Delhi.

Methodology

1. **Data Preprocessing:**
   All datasets are imported and formatted to ensure consistency. Longitude and latitude values are rounded and merged into a single dataframe.
2. **Grid and Safety Score Generation:**
   A 500m × 500m grid is created across Delhi using *Shapely* geometry. Each cell is assigned a **safety score**, computed as:

   Safety Score=1−Crime CountMaximum Crime Count\text{Safety Score} = 1 - \frac{\text{Crime Count}}{\text{Maximum Crime Count}}Safety Score=1−Maximum Crime CountCrime Count

   Higher scores indicate safer regions.

3. **Graph Construction:**
   Each grid cell acts as a node, and edges are weighted based on the average safety scores of adjacent grids.
4. **Routing Algorithm:**
   The **Dijkstra algorithm** is implemented using **NetworkX** to compute the safest path between the user's source and destination points.
5. **Visualization and Interface:**
   - **Folium** is used for map visualization.
   - **Streamlit** is used for creating the front-end interface.

  - Crime zones are color-coded (green = safe, red = unsafe), and **police stations** are displayed as blue markers for easy identification.

Model Implementation

The application workflow includes:

- User input of *start* and *destination* coordinates.
- Crime-based safety score calculation for each grid cell.
- Dijkstra-based computation of the safest route.
- Visualization of the route, safety map, and nearby police stations on an interactive map.

All results are dynamically displayed on the app's GUI, built using Streamlit.

Key Findings

- The system successfully calculates and visualizes **crime density heatmaps** and **safe route predictions**.
- The Dijkstra-based approach efficiently identifies optimal safe paths within seconds.
- The visualization clearly distinguishes between low- and high-risk zones, improving user situational awareness.
- Integration of police station data provides practical real-world utility for women's safety during travel.

The project demonstrates that **spatially informed algorithms** can significantly enhance urban safety applications and promote data-driven decision-making in smart cities.

Insights and Contributions

- Introduces a **novel safety-routing framework** focused specifically on women's mobility in Delhi.
- Combines geospatial analytics, graph theory, and data visualization in a single deployable app.
- Bridges the gap between theoretical safety research and real-world implementation.
- Encourages the use of open crime data for developing social-impact technologies.

Practical Applications

- Can be used by daily commuters to identify **safer travel routes** in Delhi.
- Enables city planners and authorities to visualize **crime-prone areas** for better policy design.
- Provides a foundation for integrating **real-time crime updates** into navigation systems.
- Offers potential scalability to other major Indian cities.

Future Work

Future extensions of this research include:

1. Integration of **real-time crime reporting APIs** for live route updates.
2. Inclusion of **machine learning-based safety prediction models**.
3. Expansion of the system to other metropolitan areas such as Mumbai, Bengaluru, and Kolkata.
4. Enhancement of the user interface for mobile deployment and multi-language accessibility.

Conclusion

The Women Safety Route App represents a practical and socially impactful use of data science and geospatial analytics. By prioritizing safety over speed in navigation, the system demonstrates how technology can empower women with actionable intelligence and safer travel options.

This research bridges the domains of **urban safety, artificial intelligence, and civic technology**, offering a sustainable model for future smart-city applications.

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